PARMAR RAILWAY

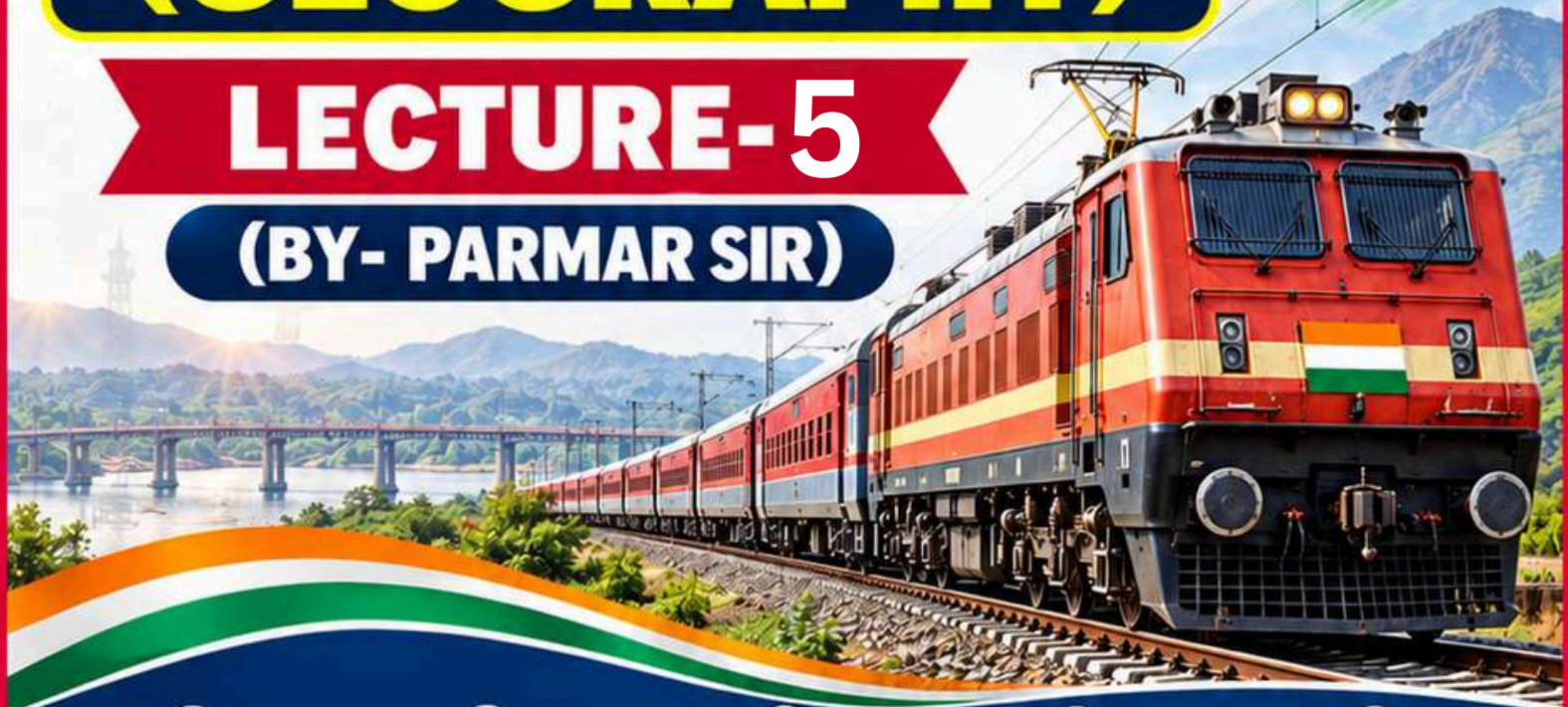
COMPLETE GK PYQ BATCH

GENERAL KNOWLEDGE

(GEOGRAPHY)

LECTURE-5

(BY- PARMAR SIR)



PHYSICAL GEOGRAPHY | INDIA GEOGRAPHY | WORLD GEOGRAPHY | ENVIRONMENT GEOGRAPHY | MAPPING CONCEPTS

LANDFORMS

- **Landforms:** Created by Geomorphic processes **(Erosional or Depositional).**
- **Agents of Erosion:** Kinetic energy causes erosion. Agents include River (fastest), Sea Water, Wind, Glacier (slow but creates massive erosion due to weight), and Ground Water.

Q1.Which of the following desert features forms due to parallel retreat of slopes through backwasting?

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (4:30 PM - 6:00 PM)

A. Playa
B. Stalagmite
C. Blowout
D. Pediplain

- **Aeolian Landforms**: Landforms created by wind, predominantly in deserts due to lack of obstacles, high wind speeds, and sand acting as an abrasive tool (Abrasion).

- **Erosional Desert Features:**
  - **Pediplain**: A broad, relatively flat rock surface formed by the joining of several pediments, leaving behind small, isolated steep-sided residual hills.
  - **Inselberg / Monadnock:** The isolated steep-sided residual hills left behind after a pediplain forms.
  - **Mushroom Rock & Pedestal Rock:** Formed when wind erodes the lower softer part of a rock more than the upper hard part.



  - **Yardang & Zeugen:** Other wind-eroded features.
  - **Playa:** Shallow depressions in deserts where water occasionally collects.
- **Depositional Desert Features:**
  - **Sand Dunes:** Sand hills formed by deposition.
    - **Barchans**: Crescent-shaped dunes with two wings pointing in the direction of the wind.
    - **Seif**: Similar to Barchan but with only one wing.
    - **Parabolic Dunes**: Crescent-shaped but the wings point in the opposite direction of the wind (due to vegetation anchoring the sides).
    - **Transverse Dunes:** Form perpendicular to the wind direction
- **Stalagmite is a groundwater depositional feature.** Peneplain is formed by river erosion (running water), unlike Pediplain which is wind-eroded.

Q2.Which of the following landforms develops when a river cuts through a ridge, forming a narrow and steep-sided passage?

NTPC UG CBT-01 2025-26 (20 Jun, 2026 (4:30 PM - 6:00 PM)

A. Plain
B. Moraines
C. Gorge
D. Dune

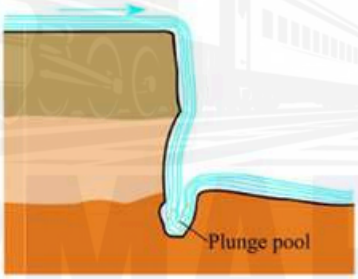
- **Dunes are formed by wind.**
- **Moraines** are formed by glaciers.
- **Plains** are formed by river deposition, not cutting (erosion).

- **River Stages and Landforms:**
  - **Youthful Stage (Mountains):** High velocity, dominant erosion. Forms V-shaped valleys, **Gorges** (steep, parallel, deep valleys), **Canyons** (step-like walls, narrow top, wide bottom), **Potholes** (small holes in the river bed), **Waterfalls**, and **Plunge Pools** (large depression where waterfall hits the ground).
  - **Mature Stage (Plains):** Starts **meandering** (twisting path due to flat terrain and Coriolis force). Forms **Alluvial Flood Plains, Braided Channels** (river splits and rejoins around deposited sediments), and **Oxbow Lakes.**
  - **Old Stage:** Slowest speed, dominant deposition. **Forms Deltas.**
- **Types of Waterfalls:**
  - **Plunge Waterfall:** Water drops vertically without touching the rock face.
  - **Cataract**: Large amount of water falling with high force.
  - **Cascade**: Water falls in a series of steps.
- **Depositional River Features:** Levees (raised banks due to flood deposition), **Alluvial Fans** (An alluvial fan is a triangle- or cone-shaped landform made of sediment (called alluvium) that is deposited where a fast-moving stream or river exits a narrow mountain canyon or hill and slows down onto a flat plain.), Flood Plains.
- **Entrenched Meander:** An erosional feature where a river meanders while cutting deep through hard rocks.

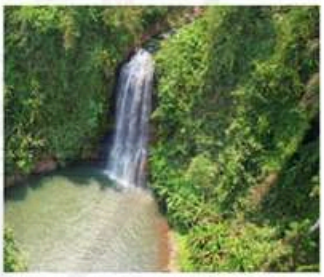
Riverine Landforms	
Erosional	Potholes, Waterfalls, Plunge Pools, V-Shaped Valley, Gorges, Canyons, River Terraces, Entrenched Meanders, Cataract Waterfall
Depositional	Flood Plains, Ox-bow lakes, Point Bar, Alluvial Fans, Levees, Braided Channel.


Pothole


V-Shaped Valley



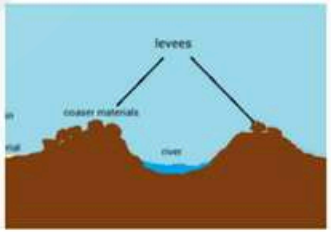
Plunge Pool


Waterfalls


River Terraces


Flood Plains


Entrenched Meanders



Levees


Alluvial Fans


Meanders


Braided Channel.


Ox-Bow lake

Q3.A plateau that is surrounded by mountains on all sides is generally referred to as a:

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (9:00 AM - 10:30 AM)

A. Lava plateau
B. Intermontane plateau
C. Dissected plateau
D. Piedmont plateau

- A plateau is a raised area of land with a flat top (tableland) and no peak.
- The **Tibetan Plateau is an Intermontane Plateau** (surrounded by the Himalayas) and is the highest plateau in the world.

Q4.Which of the following pairs is formed inside limestone caves by deposition?

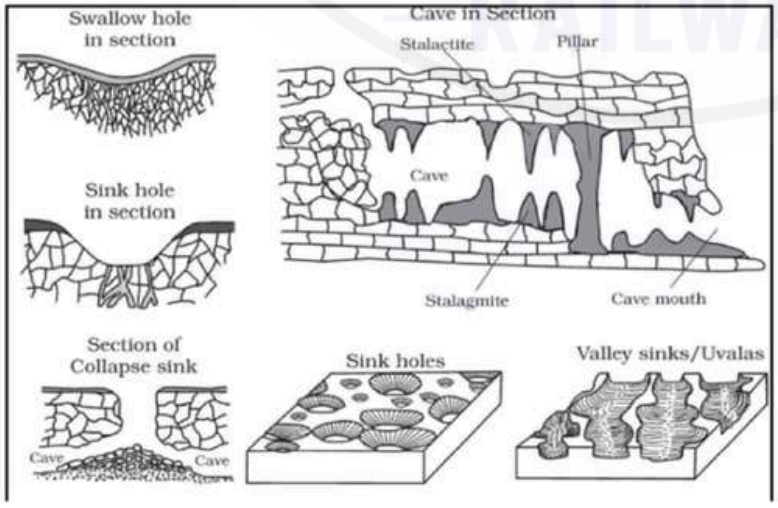
NTPC UG CBT-01 2025-26 (17 Jun, 2026 (4:30 PM - 6:00 PM)

A. Levees and meanders
B. Stalactites and stalagmites
C. Drumlins and eskers
D. Cirques and horns

- **Groundwater Landforms (Karst Topography):** Formed in areas with soft rocks like Dolomite and Limestone (e.g., Karst region near the Adriatic Sea).

- **Erosional Features:**
  - **Swallow hole** -> Sinkhole (larger).
  - **Collapse Sink:** Formed when the roof of a sinkhole collapses.
  - **Uvala**: A large valley formed when multiple sinkholes merge.
  - **Lapies** & **Limestone Pavements:** Uneven, grooved surface features.
- **Depositional Features (Inside Caves):**
  - **Stalactite**: Hangs tightly from the ceiling (Trick: 'Tite' = Tightly holds ceiling).
  - **Stalagmite**: Rises from the floor (Trick: 'Mite' = Might fall / fallen on ground).
  - **Pillar**: Formed when a stalactite and stalagmite join together.
- **An Aquifer is groundwater stored between hard rocks,** not a depositional feature. Levees/Meanders are river features. Drumlins/Eskers/Cirques/Horns are glacial features.

Karst Topography	
Erosional	Sink Hole, Dolines, Lapies, Uvalas, Limestone pavements, Swallow Hole
Depositional	Stalactite, Pillars, Stalagmite, Aquifer.





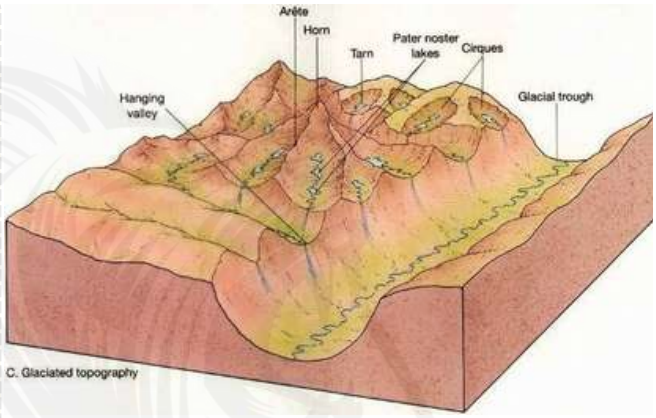
Q5. Which landform is formed when a glacier erodes a pre-existing river valley by widening, deepening, and smoothing it?

NTPC Graduation CBT-01 2025-26 (25 Mar, 2026 Shift 3)

A. Floodplain
B. Gorge
C. V-shaped valley
D. U-shaped valley

- **Floodplain, Gorge, V-shaped valley all are formed by rivers.**
- **Glaciers cause deep friction and erosion due to their massive weight.**

- **Glacial Erosional Features:**
  - **Cirque**: An armchair-shaped depression formed where glaciers originate.
  - **Tarn Lake:** A lake formed when water fills a cirque.
  - **Horn**: A sharp, pointed peak formed when multiple cirques erode backward into a mountain (e.g., Matterhorn).
  - **Arete**: A sharp, jagged ridge separating two adjacent cirques.



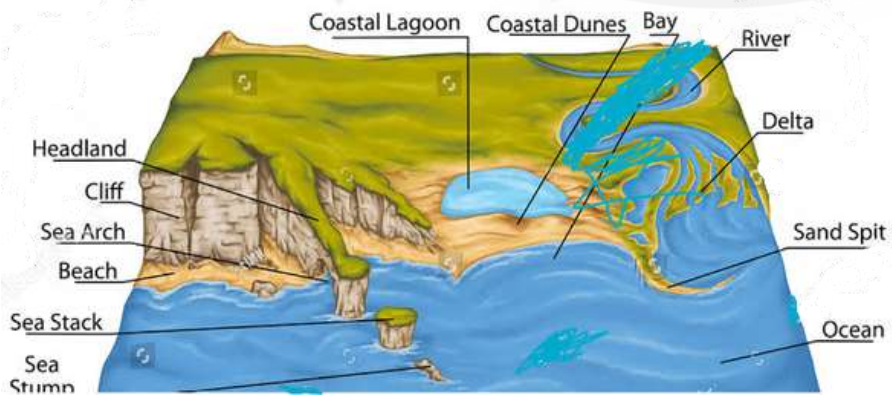
  - **U-shaped Valley**: Deep, wide valley carved by a moving glacier.
  - **Hangings Valley:** A tributary valley that sits higher than the main U-shaped valley.
  - **Fjord**: A glacial valley filled with seawater.
- **Glacial Depositional Features:**
  - **Moraines**: Glacial debris. (Lateral moraines are on the sides, Medial moraines are in the middle).
  - **Drumlins**: Oval, drum-shaped hills of glacial till (also called **'Basket of Eggs' topography).**
  - **Eskers**: Long, winding ridges of gravel and sand (Trick: Ris-ris kar 'Esker' ban gaya).
  - **Outwash Plains:** Vast plains formed by glacial meltwater deposition. **(Trick: Morena people play 'Drums' (Drumlins) and 'Eskers' in the 'Outwash Plains').**

Q6. Sand dunes that form long ridges parallel to the coastline are most common _____.

RRB NTPC UG CBT-01 2024 (09 Sept, 2025 Shift 2)

A. along low sedimentary coasts
B. in estuarine regions only
C. along high rocky coasts
D. along tectonic plate boundaries

- **This refers to features formed by Sea Water. Depositional features** are prominent on the East Coast (low sedimentary coast), while erosional features dominate the West Coast.



- **Sea Water Erosional Features:**
  - **Cliff**: Steep rock faces on the coast.
  - **Sea Cave:** A hollowed-out section in a cliff.
  - **Sea Arch:** Formed when a cave is eroded entirely through a headland.

- Sea Stack: Isolated pillars of rock left standing when a sea arch collapses.

- **Sea Water Depositional Features:**
  - **Beach**: Shoreline covered with sand.
  - **Sand Spit**: A narrow strip of sand jutting out into the sea.
  - **Bar**: A ridge of sand parallel to the coast.
  - **Tombolo**: A ridge of sand connecting the mainland to an island.

Q7. The western coastal plains are an example of:

RRB NTPC UG CBT-01 2024 (28 Aug, 2025 Shift 3)

A. tectonic coast
B. submerged coast
C. volcanic coast
D. emergent coast

- The **Western coastal plain is submerged (narrow), while the Eastern coastal plain is emergent (wider and retreating).**

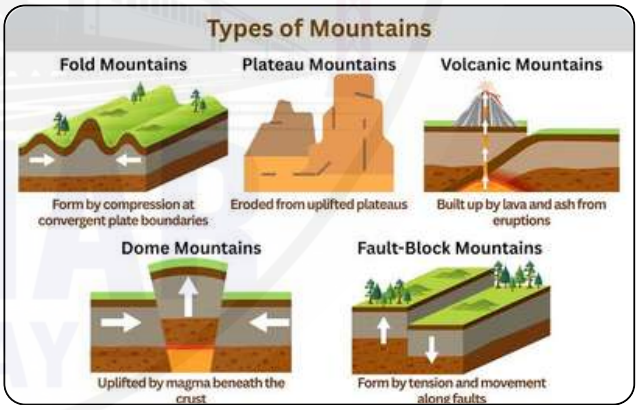


Q8. Which type of mountain range is formed by the uplifted folded sedimentary rocks?

RRB NTPC UG CBT-01 2024 (21 Aug, 2025 Shift 1)

A. Volcanic Mountains
B. Block Mountains
C. Fold Mountains
D. Residual Mountains

- **Geomorphic Processes:**
  - **Endogenic (Internal) Forces:** Diastrophism (building processes like mountains) and sudden events (Volcanoes, Earthquakes).
    - **Orogenic**: Mountain building (e.g., Fold mountains).
    - **Epeirogenic**: Continent building (uplifting a large part of the earth's crust).
  - **Exogenic (External) Forces:** Denudation (destroying/eroding processes). Powered by solar energy.



- **Fold Mountains:** Formed by folding of sedimentary rocks due to compressive forces.
  - **Young Fold Mountains:** Himalayas, Andes, Rockies, Alps.
  - **Old Fold Mountains:** Aravallis, Appalachians.
- **Residual Mountains:** When Exogenic forces wear down an old fold mountain significantly (like the Aravallis, now very low). The Raisina Hill in Delhi is a remnant of the Aravallis.
- **Block Mountains:** Formed by faulting. **The uplifted block is a Horst (Block Mountain)** and the **sunken block is a Graben (Rift Valley). Example**: Vindhya and Satpura (Horsts), Narmada flows in the Rift Valley (Graben) between them.

Q9.Which mountain is located in South America?

A. Rockies
B. Alps
C. Vosges
D. Andes



RRB NTPC UG CBT-01 2024 (20 Aug, 2025 Shift 1)

- **Rockies**: North America (Young Fold). Highest peak: Mount Denali (renamed Mount McKinley).
- **Alps**: Europe (Young Fold). Highest peak: Mont Blanc.
- **Vosges**: France (Block Mountain).
- **Appalachians**: North America (Old Fold Mountain).
- **Andes**: South America. It is the longest mountain range in the world. Highest peak: Mount Aconcagua.

Q10.Why does the eastern coastal plain have fewer ports and harbours compared to the western coastal plain?

RRB NTPC UG CBT-01 2024 (19 Aug, 2025 Shift 3)

A. It is closer to the Equator
B. It has more tourist beaches
C. It is an emergent coast
D. It receives more rainfall



- **The Western coast** (submerged) is naturally deeper and more favorable for port formation. **The Eastern coast** (emergent) is shallower. Most ports on the east coast are artificial.

- **Major Ports of India (13 Total – 12 Public, 1 Private):**
  - **West Coast:** Kandla (Deendayal Upadhyay, a tidal port), Mumbai, JNPT (Nhava Sheva, largest container & satellite port), Mormugao (Goa, on Zuari estuary, 1st in iron export), New Mangalore (Karnataka), Kochi (Kerala, 'Queen of the Arabian Sea').
  - **East Coast:** Ennore (Kamarajar, 1st major corporatized port), Chennai, Tuticorin (Tamil Nadu has 3 ports), Visakhapatnam (Deepest landlocked port), Paradip (Odisha), Kolkata (Diamond Harbour, oldest port).

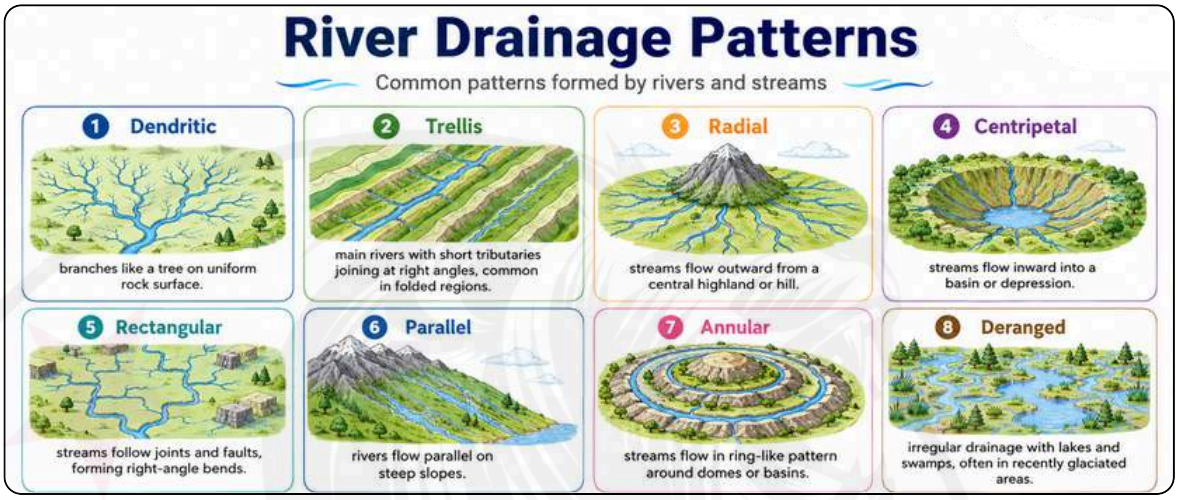
- **Tides (Jwar-Bhata)**
  - Tides are caused by the Gravitational pull of the Sun and Moon and the Centrifugal force of the Earth.
    - **Spring Tide:** High amplitude tides occurring when the Sun, Moon, and Earth are in a straight line (Syzygy position).
    - **Neap Tide:** Low amplitude tides occurring when the Sun, Earth, and Moon form a 90-degree angle (Quadrature position).

- **Diurnal Tide:** 1 High and 1 Low tide in 24 hours and 50 minutes (Interval of 12 hours 25 minutes).
- **Semi-Diurnal Tide:** 2 High and 2 Low tides in a day.

Q11. Which drainage pattern resembles the branches of a tree and is common in rivers of the northern plains?

RRB NTPC UG CBT-01 2024 (13 Aug, 2025 Shift 3)

A. Dendritic
B. Trellis
C. Centripetal
D. Radial

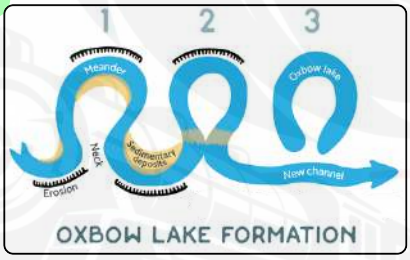


Q12. Which of the following is an impact of river meandering in plains?

A. Formation of waterfalls

RRB NTPC UG CBT-01 2024 (13 Aug, 2025 Shift 2)

B. Formation of oxbow lakes
C. Formation of rift valleys
D. Formation of rapids



- **Meandering** is caused by low gradients in plains and the Coriolis force.
- **When a meander loop cuts off from the main river, an Ox-bow lake is formed.**

Q13. The crescent-shaped sand dunes found in the Thar Desert are called:

RRB NTPC Graduation CBT-01 2024 (20 Jun, 2025 Shift 1)

A. Barchans
B. Ergs
C. Inselbergs
D. Yardangs

Earth's Interior & Plate Tectonics

Q14. High sea waves triggered by an undersea earthquake are called:

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (4:30 PM - 6:00 PM)

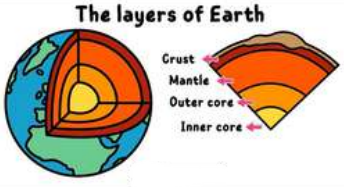
A. Monsoon winds
B. Tsunamis
C. Land depressions
D. Glacial surges

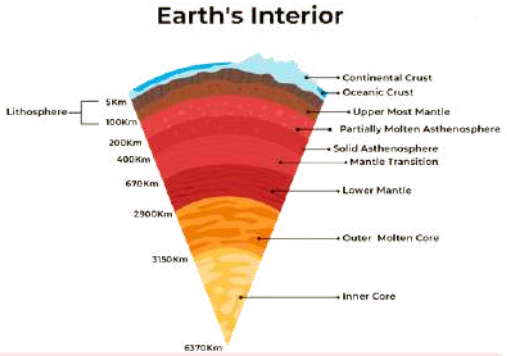
- **Tsunami** is a Japanese term meaning **"Harbour waves".**
- They are specifically caused by undersea earthquakes, not weather patterns (like cyclones).

Q15. The asthenosphere plays an important role in plate movements because:

NTPC UG CBT-01 2025-26 (15 Jun, 2026 (12:45 PM - 2:15 PM)

A. It is partially molten and allows plates to move
B. It is made of water bodies
C. It is perfectly solid and rigid
D. It forms the Earth's magnetic field





- **Earth's Layers:**
  - **Crust**: 1% by volume, <1% by mass. Uppermost layer, Up to 100 km depth..
    - **Continental Crust:** Thicker, made of SIAL (Silica + Aluminum).
    - **Oceanic Crust:** Thinner but denser (heavier), made of SIMA (Silica + Magnesium).
  - **Mantle**: 84% by volume, 68% by mass. Up to 2900 km depth. Made of SIMA.
  - **Core**: 15% by volume, 31% by mass. Up to 6400 km depth. Made of NIFE (Nickel + Iron).
    - **Outer Core:** Liquid (causes Earth's magnetic field).
    - **Inner Core:** Solid (due to extremely high pressure).
  - **Lithosphere**: Crust + Upper solid part of the mantle (up to 200 km deep). These form tectonic plates.
    - **Asthenosphere** (Durbal Mandal): Below the lithosphere (up to 400 km). It is semi-molten magma (weak zone). The lithospheric plates slide over this layer.

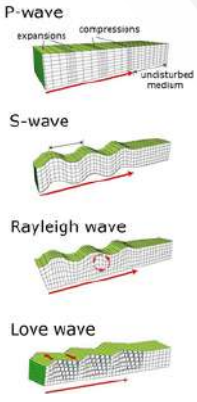
Q16. The zone on Earth where no S-waves are detected is known as:

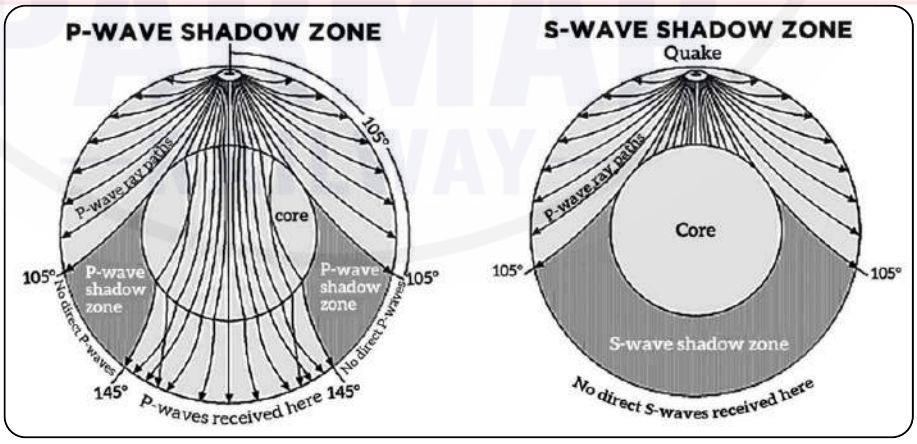
NTPC UG CBT-01 2025-26 (14 Jun, 2026 (12:45 PM - 2:15 PM)

A. Epicentral gap
B. Active zone
C. Shadow zone
D. Seismic belt

- **Earthquakes**: Originate at the Hypocenter (Focus). The point directly above on the surface is the Epicenter (where maximum destruction occurs).

- **Seismic Waves:**
  - **Body Waves (Travel through the earth):**
    - **P-Waves (Primary):** Fastest, reach the seismograph first. Travel through Solid, Liquid, Gas. Longitudinal nature (causes stretching and squeezing).
    - **S-Waves (Secondary):** Travel only through Solids. Transverse nature (creates crests and troughs).
  - **Surface Waves (L-waves & Rayleigh waves)**: Travel on the surface, most destructive. (Rayleigh waves roll, Love waves move side-to-side).





- **Shadow Zones:** Areas where seismographs do not detect waves.
  - **Because the Outer Core is liquid,** S-waves cannot pass through it. Their shadow zone is huge (from 105° to 105° on the other side, covering ~40% of the earth).
  - **P-waves pass through the liquid core** but refract (slow down), creating a smaller shadow zone from 105° to 145°.

Q17. Which seismic waves are the fastest and travel through both solids and liquids inside the Earth?

NTPC UG CBT-01 2025-26 (09 May, 2026 (12:45 PM - 2:15 PM)

A. Rayleigh waves
B. P waves
C. S waves
D. Surface waves

Q18. Which layer of the Earth plays the primary role in generating the planet's magnetic field?

NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 1)

A. Crust
B. Mantle
C. Outer core
D. Inner core

Q19. Which of the following instruments is used to measure the intensity of earthquakes?

NTPC Graduation CBT-01 2025-26 (25 Mar, 2026 Shift 3)

A. Tiltmeter
B. Seismograph
C. Anemometer
D. Barometer

- **Intensity** is measured by the Mercalli scale (1-12 limit).
- **Magnitude** is measured by the Richter scale (0-10 generally, but it is limitless).
- **Anemometer** measures wind speed.
- **Barometer** measures pressure.

Q20. At approximately what depth inside the Earth is the boundary between the mantle and the core located?

NTPC Graduation CBT-01 2025-26 (25 Mar, 2026 Shift 1)

A. 4,900 km
B. 1,900 km
C. 3,900 km
D. 2,900 km

- **Discontinuities Trick (Koi Mujhe Red Gulab La Do):**
  - **Conrad**: Upper Crust / Lower Crust.
  - **Mohorovicic** (Moho): Crust / Mantle.
  - **Repetti**: Upper Mantle / Lower Mantle.
  - **Gutenberg**: Mantle / Core (at 2900 km).
  - **Lehmann**: Outer Core / Inner Core.

Q21. Which of the following processes is/are primarily responsible for the formation of fold mountains such as the Himalayas?

NTPC Graduation CBT-01 2025-26 (24 Mar, 2026 Shift 1)

A. Wind deposition
B. Erosion by rivers
C. Volcanic eruptions
D. Plate collision

- Himalayas formed due to the collision (Convergent boundary) of the Indian Plate and the Eurasian Plate, closing the Tethys Sea.

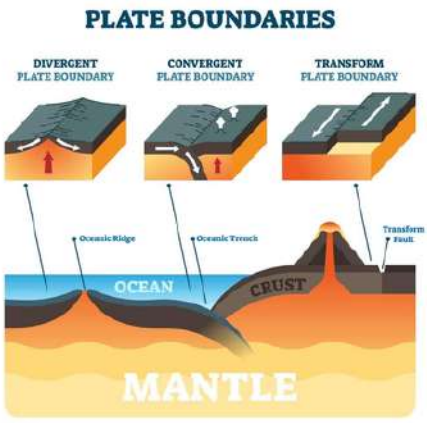
Q22. What happens to tectonic plates at divergent boundaries?

RRB NTPC UG CBT-01 2024 (14 Aug, 2025 Shift 2)

A. They stay in one place.
B. They move away from each other.
C. They slide past one another.
D. They move towards each other.

- **Tectonic Plates: 7 Major** (Antarctica, Eurasian, African, N. American, S. American, Pacific, Indo-Australian) and **many minor (Nazca, Cocos, Arabian, Philippine, etc.).**

- **Convergent Boundary:** Plates move towards each other. The denser plate (Oceanic) subducts under the lighter plate (Continental) into the asthenosphere and melts. **Highly Destructive. Forms Subduction Zones and Fold Mountains.**
- **Divergent Boundary:** Plates move away. Magma rises and creates new crust. **Highly Constructive. Forms Mid-Oceanic Ridges** (**e.g., Mid-Atlantic Ridge**, the longest landform on earth). Leads to Seafloor Spreading (Discovered by Harry Hess in 1962).



- **Transform Boundary:** Plates slide past each other (Conservative boundary).

- **Arthur Holmes' Convection Current Theory (1930):** Explains plate movement. Magma in the asthenosphere moves due to convection currents powered by residual heat and radioactive decay, dragging the plates above. (Alfred Wegener's 1912 Continental Drift Theory wrongly blamed polar fleeing and tidal forces).

Geomorphic Processess

Q23. The peeling of curved sheets from massive rocks due to temperature changes is known as:

NTPC UG CBT-01 2025-26 (17 Jun, 2026 (4:30 PM - 6:00 PM)

A. Exfoliation
B. Carbonation
C. Salt crystallisation
D. Abrasion

- **Weathering**: The breaking down of rocks in-situ (without movement).
  - **Mechanical / Physical Weathering:** Exfoliation (due to thermal expansion/contraction), Frost Wedging (water freezing in cracks and expanding), Abrasion.
  - **Biological Weathering:** Roots, rodents breaking rocks.
  - **Chemical Weathering:** Carbonation, Hydration, Oxidation, Reduction, Salt Crystallization.

Q24. Which process mainly helps in the formation of V-shaped valleys in mountainous regions?

NTPC UG CBT-01 2025-26 (14 Jun, 2026 (9:00 AM - 10:30 AM)

A. River erosion
B. Wave action
C. Glacier movement
D. Wind erosion

Q25. Which of the following statements about volcanism is most accurate in the context of geomorphic processes?

NTPC UG CBT-01 2025-26 (09 May, 2026 (12:45 PM - 2:15 PM)

A. It mainly builds or elevates parts of the crust
B. It only lowers existing landforms
C. It is controlled by rainfall and temperature
D. It requires running water to operate

- **Volcanism** is an Endogenic process that creates new landforms by depositing magma/lava.

Q26. Which of the following types of weathering is primarily responsible for the formation of karst landscapes?

NTPC Graduation CBT-01 2025-26 (24 Mar, 2026 Shift 3)

A. Mechanical weathering
B. Biological weathering
C. Chemical weathering
D. Thermal weathering

Q27. Landslides due to mining and land subsidence are commonly seen in:

RRB NTPC UG CBT-01 2024 (19 Aug, 2025 Shift 3)

A. Assam and West Bengal
B. Himachal Pradesh and Uttarakhand
C. Rajasthan and Haryana
D. Jharkhand and Chhattisgarh

- **Denudation** is the overall geological process that wears down and lowers the Earth's surface through the combined action of **weathering, mass wasting (mass movement), and erosion.**
  - **Mass Movement**: Downhill movement of rock/soil due to gravity. Weathering aids it but is not a prerequisite.
    - **Rapid Movements:** Landslide, Avalanche, Earth flow, Mudflow.
    - **Slow Movements:** Creep (very slow), Solifluction (water-saturated soil flowing downhill).
  - **Slump**: Backward rotational movement of debris.

Ocean and Continents

Q28. Which continent is known as the 'Island Continent'?

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (4:30 PM - 6:00 PM)

A. Europe
B. Africa
C. South America
D. Australia

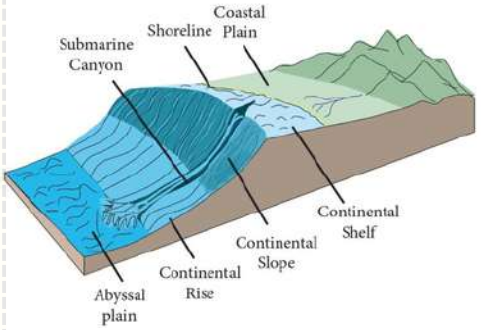
Q29. Which of the following is characterised as an interconnected chain of submerged mountains, with a central rift zone of intense volcanic activity?

NTPC UG CBT-01 2025-26 (19 Jun, 2026 (4:30 PM - 6:00 PM)

A. Continental slope
B. Mid-oceanic ridges
C. Deep-ocean trenches
D. Abyssal plains

- **Mid-oceanic ridges** are continuous, underwater volcanic mountain chains that **form at divergent plate boundaries** where tectonic plates pull apart and magma rises from the Earth's mantle to create new oceanic crust
  - **Seafloor spreading** is a geological process where tectonic plates pull apart at mid-ocean ridges, allowing magma from the mantle to rise, cool, and form new oceanic crust.
    - **This theory was proposed by Harry Hess** (and further developed with contributors like Robert Dietz and mapping by Marie Tharp) in the early 1960s

- **Ocean Floor Relief Features:**
  - **Continental Shelf:** Shallowest part, gradient 1° or less. Major area for oil/mineral extraction.
  - **Continental Slope:** Steep drop.
  - **Continental Rise:** Gentle slope leading to the deep ocean.
  - **Abyssal Plain (Deep Sea Plain):** Flat ocean floor. Features Seamounts (pointed underwater mountains) and Guyots (flat-topped seamounts).



Q30. Which continent has the highest average elevation due to its thick ice sheets?

NTPC UG CBT-01 2025-26 (16 Jun, 2026 (4:30 PM - 6:00 PM)

A. Africa
B. Europe
C. Antarctica
D. South America

- **Africa is the Continent of Continents (most countries).** Highest peak: Mount Kilimanjaro (Volcanic, in Tanzania). Atlas Mountains are in Northern Africa.

Q31. What does the term 'halocline' refer to in oceanography?

RRB NTPC UG CBT-01 2024 (01 Sept, 2025 Shift 3)

A. A layer with a rapid change in pressure
B. A layer with a rapid change in salinity
C. A zone with the highest oxygen concentration
D. A layer with a rapid change in temperature

- **Halo** = Salt. Salinity increases rapidly downwards in the halocline.
- **Thermocline** = Rapid decrease in temperature downwards.

Q32. How many deep trenches are there in the Indian Ocean?

RRB NTPC UG CBT-01 2024 (20 Aug, 2025 Shift 3)

A. Eight
B. Nine
C. Seven
D. Six

- **Pacific Ocean has the most trenches (32).**
- **Ocean Size Order (Trick: PAISA):** Pacific > Atlantic > Indian > Southern > Arctic.
- **Deepest trench globally:** Mariana Trench (11,000m) in the Pacific.
- **Deepest trench in the Indian Ocean:** Sunda Trench (Java Trench), roughly 7900m deep.

Cyclone & Ocean Currents

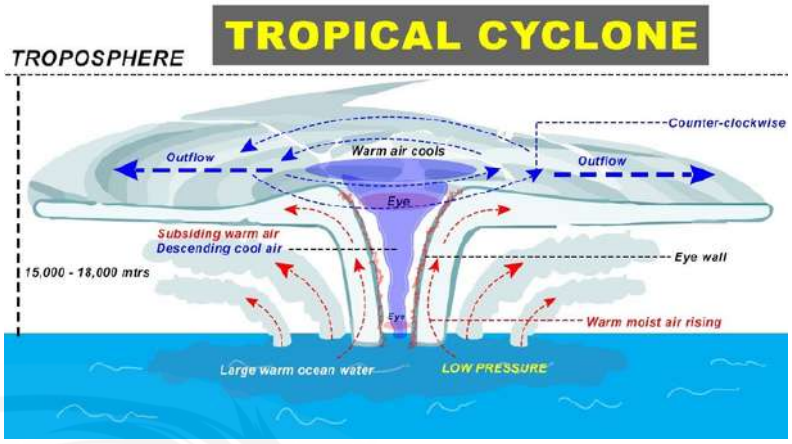
Q33. Tropical cyclones form mainly due to:

NTPC UG CBT-01 2025-26 (17 Jun, 2026 (9:00 AM - 10:30 AM)

A. Fast-moving cold fronts
B. Rising continental winds
C. Low-pressure systems over warm ocean water
D. High-pressure systems over land

- **Conditions for Tropical Cyclones:**
  - Large sea surface with temperature > 27°C.
  - Pre-existing weak Low-Pressure area.
  - Presence of Coriolis force (to spin the cyclone).

- They **form between the Tropics. Inward spiraling winds.**
- **Eye of the Cyclone:** The calm, cloudless center.
- **Clouds**: **Formed around the eye. Specifically, Cumulonimbus clouds** (massive vertical development, anvil-shaped top, bring heavy rain).



- **Cloud Types Overview:**
  - **Low (Up to 4km):** Stratus, Nimbostratus (Dark grey, block sunlight, heavy rain).
  - **Middle (4-8km):** Cumulus (Flat base, cauliflower/cotton wool shape).
  - **High (8-12km):** Cirrus (Feathery, icy crystals).
- **Cyclone Spin Direction:**
  - **Northern Hemisphere:** Anti-clockwise (Cyclones) / Clockwise (Anti-cyclones).
  - **Southern Hemisphere:** Clockwise (Cyclones) / Anti-clockwise (Anti-cyclones).
- **Cyclones form around Low Pressure. Anti-cyclones form around High Pressure** (Clear, sunny weather).
- **Temperate Cyclones:** Form in mid-latitudes due to Frontogenesis (meeting of Warm and Cold fronts). Can form over land and sea. They move West to East (Tropical cyclones move East to West).
- **Occluded Front:** When a cold front overtakes a warm front, lifting the warm air mass entirely off the ground.

Q34. Which ocean current is primarily responsible for moderating the climate of Western Europe by transporting warm water from the tropics?

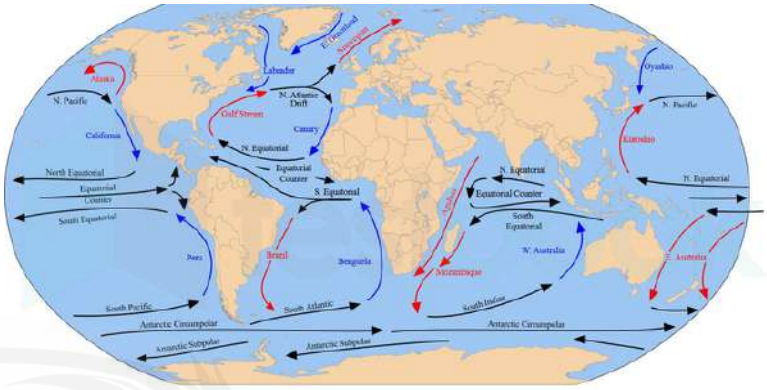
NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 2)

A. Humboldt Current
B. North Atlantic Drift
C. Oyashio Current
D. Canary Current

- **Ocean Currents:** Driven by Insolation (heating), Density differences, Gravity, and Coriolis force.

- **Cold currents create deserts** on the Western coasts of continents because cold air holds less moisture, leading to no rain.
- **Fishing Zones:** Form where Warm and Cold currents meet (e.g., Labrador + Gulf Stream near Newfoundland). However, these areas are foggy, making them bad for harboring ships.
- **Gyre**: A large system of circulating ocean currents.
- **Equatorial Counter Current:** Flows backwards (West to East) between the North and South Equatorial currents.
- **The North Atlantic Drift** is an extension of the warm Gulf Stream, keeping Western Europe (like Norway) ice-free.

- **Cold Currents (Trick: BP Golf California):**
  - Benguela (Africa Coast)
  - Peru / Humboldt (Creates Atacama Desert)
  - Greenland
  - Okhotsk (Oyashio)
  - Labrador
  - Falkland
  - West Australian
  - Canary (Creates Sahara Desert)
  - California



Q35. Canary Current occurs in which of the following oceans?

NTPC Graduation CBT-01 2025-26 (17 Mar, 2026 Shift 3)

A. Pacific Ocean
B. Atlantic Ocean
C. Indian Ocean
D. Southern Ocean

Q36. Where do tropical cyclones occur?

RRB NTPC UG CBT-01 2024 (08 Sept, 2025 Shift 2)

A. In very high altitude deserts
B. Near the equatorial belt
C. In tropical low-pressure areas
D. In mid-latitude regions

Q37. What is the primary destructive force of tropical storms occurring in coastal regions?

NTPC Graduation CBT-01 2025-26 (17 Mar, 2026 Shift 2)

A. Continuous rainfall
B. Storm surge
C. High temperature
D. Low atmospheric pressure

- **Tropical cyclones derive their energy** from warm seawater. When they hit land (coast), they dump this energy and water as a massive "Storm Surge", causing immense destruction. Once inland, deprived of moisture, they weaken rapidly.

Atmosphere

Q38. Which of the following does NOT influence the horizontal distribution of air pressure?

NTPC UG CBT-01 2025-26 (18 Jun, 2026 (9:00 AM - 10:30 AM)

A. Presence of water vapour
B. Ocean salinity
C. Air temperature
D. Earth's rotation

Q39. Which river forms a fertile region within parts of the Great Indian Desert?

NTPC UG CBT-01 2025-26 (16 Jun, 2026 (9:00 AM - 10:30 AM)

A. Sabarmati River
B. Narmada River
C. Beas River
D. Ghaggar River

Rocks & Volcanos

Q40. Which of the following provides evidence that the Great Indian Desert was once under the sea during the Mesozoic era?

NTPC UG CBT-01 2025-26 (16 Jun, 2026 (4:30 PM - 6:00 PM)

A. Presence of black soil
B. Volcanic ash layers
C. Marine deposits and wood fossils
D. Glacial deposits

Q41. Which region is known as the "Ring of Fire" due to frequent volcanic activity?

NTPC Graduation CBT-01 2025-26 (19 Mar, 2026 Shift 1)

A. East African Rift
B. Mid Atlantic Ridge
C. Pacific Ocean Rim
D. Indian Ocean Floor

- **The Ring of Fire is a horseshoe**-shaped area in the Pacific Ocean.
- **It has convergent plate boundaries.** ~90% of volcanoes and ~70% of earthquakes occur here.

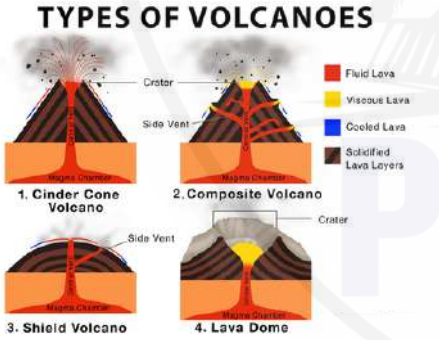


Q42. Which volcano type is also known as a stratovolcano due to its layered structure?

NTPC Graduation CBT-01 2025-26 (17 Mar, 2026 Shift 3)

A. Shield volcano
B. Composite volcano
C. Cinder cone volcano
D. Fissure volcano

- **Shield Volcano:** Highly fluid (basic) lava, spreads far, creates gentle slopes (e.g., Flood Basalt Provinces).
- **Composite (Strato) Volcano**: Thick, viscous (acidic/high silica) lava. Creates steep, layered cones.
  - **Caldera**: The most explosive and destructive volcano. It collapses into its own empty magma chamber after erupting.
  - **Crater**: The bowl-shaped depression at the top of the volcano vent.



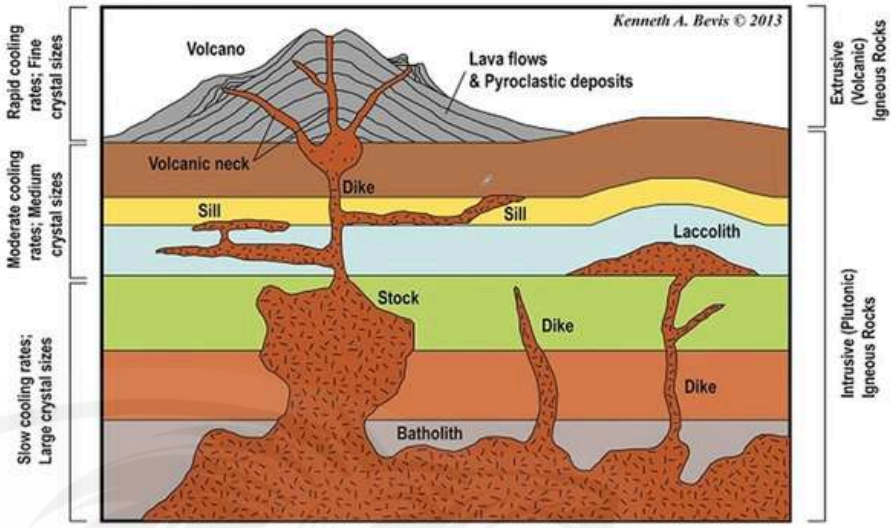
Q43. What is the name given to a saucer-shaped intrusive igneous body that is concave to the sky?

RRB NTPC UG CBT-01 2024 (02 Sept, 2025 Shift 3)

A. Lapolith
B. Sill
C. Phacolith
D. Batholith

- **Intrusive Igneous (Plutonic) Rocks: Formed when magma solidifies inside the earth.**
  - **Batholith**: Huge solidified magma chamber at the base.
  - **Dyke**: Vertical solidification.
  - **Sill**: Horizontal solidification.
  - **Laccolith**: Dome-shaped, connected by a conduit (pipe-like structure).

- **Phacolith**: Wavy mass formed in anticlines (crests) and synclines (troughs).
- **Lopolith**: Saucer-shaped, concave to the sky.



TYPES OF ROCKS:

- **Igneous Rocks:** Primary rocks formed by cooling magma/lava. **Examples**: Gabbro, Granite, Basalt, Pumice.
- **Sedimentary Rocks:** Formed by compaction of eroded sediments layer by layer (**Lithification**). **Contains fossils**. **Types**: Mechanically formed, Chemically formed, Organically formed. **Examples**: Sandstone, Limestone, Chalk.
- **Metamorphic Rocks:** Formed when Igneous or Sedimentary rocks undergo intense Pressure, Volume, and Temperature (PVT) changes without fully melting. **Examples**: Marble, Quartze.
  - **Dynamic Metamorphism:** Metamorphism without appreciable chemical change.
  - **Foliation/Lineation:** Minerals arranged in lines.
  - **Banding:** Alternating light and dark shades.
  - **Transformations**: Limestone -> Marble, Sandstone -> Quartzite, Granite -> Gneiss, Clay/Shale -> Slate.
- **Rock Cycle:** The continuous transformation of one rock type into another.

Q44. Port Victoria, where the First Training Squadron completed its visit, serves as the capital port of which African island nation?

NTPC Graduation CBT-01 2025-26 (27 Mar, 2026 Shift 2)

A. Madagascar
B. Comoros
C. Seychelles
D. Mauritius

- **Madagascar Capital:** Antananarivo.
- **Comoros Capital:** Moroni.
- **Mauritius Capital:** Port Louis.

Q45. Mont Blanc is the highest peak of which mountain range?

RRB NTPC UG CBT-01 2024 (19 Aug, 2025 Shift 3)

A. Alps
B. Himalayas
C. Western Ghats
D. Andes

Q46. Which geographical feature separates India from Sri Lanka?

RRB NTPC UG CBT-01 2024 (08 Aug, 2025 Shift 1)

A. Strait of Hormuz
B. Gulf of Kutch
C. Palk Strait and Gulf of Mannar
D. Gulf of Aden

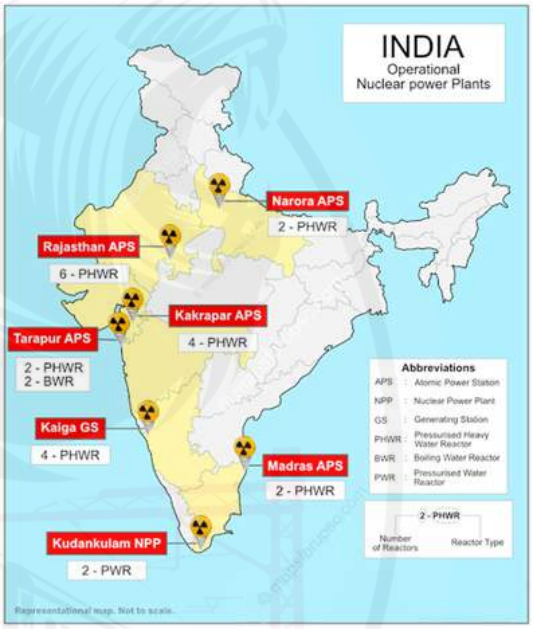
- A **strait connects two large water bodies** (Bay of Bengal and Arabian Sea) and separates two landmasses (India and Sri Lanka).
- **Strait of Hormuz connects the Persian Gulf to the Gulf of Oman.**



Energy Resources

Q47. Identify the southernmost atomic/nuclear power station of India from the following options. NTPC Graduation CBT-01 2025-26 (18 Mar, 2026 Shift 3)

A. Kakrapar Atomic Power Station
B. Narora Atomic Power Station
C. Madras Atomic Power Station
D. Kudankulam Nuclear Power Station



- **Narora**: UP
- **Kakrapar**: Gujarat
- **Tarapur**: Maharashtra (Oldest, 1969. Apsara reactor installed here).
- **Kaiga**: Karnataka
- **Madras** (Kalpakkam) & Kudankulam: **Tamil Nadu** (Kudankulam is the southernmost).
- **Nuclear Reactor basics:** Uses Uranium. Controlled by Moderators (Heavy Water, Graphite) and Control Rods (Cadmium, Boron). Coolant is also Heavy water.

Condensation & Precipitation

Q48. Which of the following is responsible for causing convectional rainfall? RRB NTPC UG CBT-01 2024 (22 Aug, 2025 Shift 1)

A. Cyclonic movements
B. Intense surface heating causing air to rise
C. Mountains forcing air to rise
D. Cold and warm air masses meeting

- **Intense heating** causes air to rise, form clouds, and cause convectional rainfall.
- **Mountain barriers** cause Orographic (Relief) rainfall.
- **Cold/Warm fronts cause** Frontal (Cyclonic) rainfall in temperate regions.

Q49. What type of rainfall occurs when air is forced to rise over a mountain barrier? RRB NTPC UG CBT-01 2024 (11 Aug, 2025 Shift 2)

A. Cyclonic rainfall
B. Orographic rainfall
C. Convectional rainfall
D. Frontal rainfall

Q50. Convectional rainfall is common in which part of India during the pre-monsoon period?

RRB NTPC Graduation CBT-01 2024 (19 Jun, 2025 Shift 3)

A. Chhota Nagpur Plateau
B. Eastern Himalayas
C. Western Ghats
D. Kerala coast



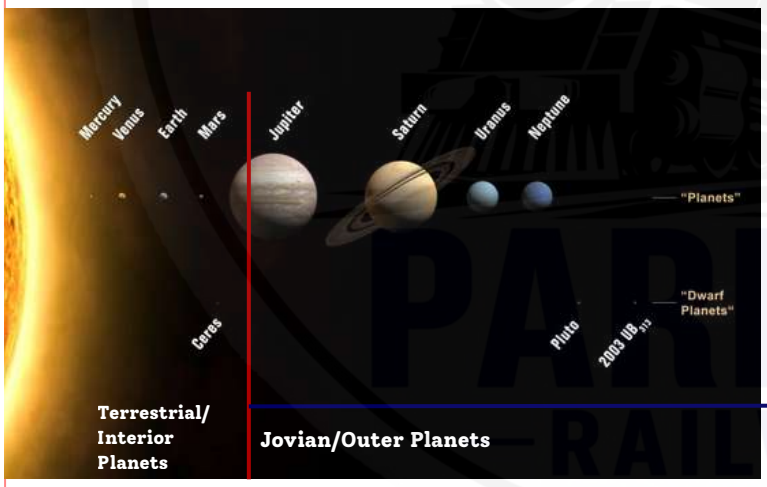
Solar System

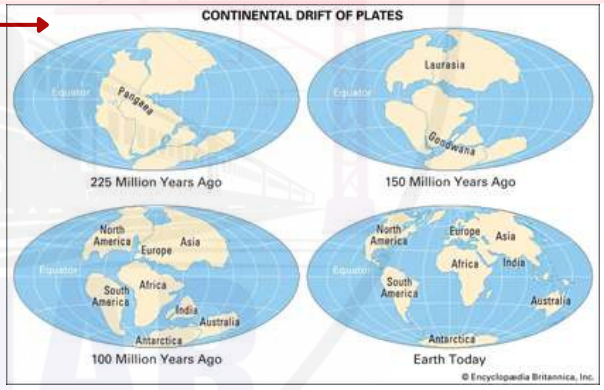
Q51. Which one of the following hypotheses/theories explains the formation of the solar system from a rotating cloud of gas and dust?

RRB NTPC UG CBT-01 2024 (18 Aug, 2025 Shift 1)

A. Big Bang Theory
B. Nebular Hypothesis
C. Continental Drift Theory
D. Plate Tectonics Theory

- **Nebular Hypothesis Proposed by Immanuel Kant and Pierre-Simon Laplace**. The rotating cloud of dust and gas is called a "Nebula".
- **Big Bang Theory:** Explains the origin of the Universe. **Given by Georges Lemaître in 1927.**
- **Continental Drift Theory:** Explains movement of continents (Pangea splitting into Laurasia and Gondwanaland). Given by Alfred Wegener in 1912.
  - **Evidence**: Jig-saw fit, Fossils, Placer (Gold) deposits in Africa and S. America, Tillite (Glacial) deposits.





Asteroid Belt: Between Mars and Jupiter. Ceres is a dwarf planet here.

- **Terrestrial/Inner Planets (Rocky, High Density, No/Few Moons):**
  - **Mercury** (**shortest revolution:** 88 days),
  - **Venus** (Earth's Twin, Hottest due to H2SO4 clouds, Brightest, Morning/Evening star, **Longest rotation:** 243 days. **Rotates East to West**, highly tilted "rolling planet"),
  - **Earth** (Densest planet, Blue planet. 71% water. Only 2.8% is fresh water. 68% of fresh water is in glaciers/ice caps. Fresh snow has the highest "**Albedo**" - light reflectivity),
  - **Mars** (Red due to Iron Oxide. **Two moons:** Phobos, Deimos).

- **Jovian/Outer Planets (Gaseous, Low Density, Big, Many Moons, Rings):**
  - **Jupiter** (**Shortest rotation:** 9h 55m. **Galilean moons:** Io, Europa, Ganymede (largest in solar system), Callisto),
  - **Saturn** (Lowest density – less than water. Rings discovered by Cassini, gaps by Huygens. **Largest moon:** Titan),
  - **Uranus** (Rotates East to West. **Discovered by William Herschel**),
  - **Neptune** (**Longest revolution:** 164 years. **Discovered by Johann Galle and Urbain Le Verrier**).
    - **Uranus and Neptune are "Ice Giants".**



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